

IMAGE PROCESSING USING MATLAB:

Image Processing Techniques



Dr Anil Kumar Maini is former director, Laser Science and Technology Centre, a premier laser and optoelectronics R&D laboratory of DRDO of Ministry of Defence



Varsha Agrawal is a senior scientist with Laser Science and Technology Centre (LASTEC), a premier R&D lab of DRDO

Image processing covers a wide and diverse array of techniques and algorithms. Fundamental processes underlying these techniques include sharpening, noise removal, deblurring, edge extraction, binarisation, contrast enhancement, and object segmentation and labelling.

Sharpening enhances the edges and fine details of an image for viewing by human beings. It increases the contrast between bright and dark regions to bring out image features. Basically, sharpening involves application of a high-pass filter to an image.

Noise removal techniques reduce the amount of noise in an image before it is processed any further. It is necessary in image processing and image interpretation so as to acquire useful information. Images from both digital cameras as well as conventional film cameras pick up noise from a variety of sources. These noise sources include salt-and-pepper noise (sparse light and dark disturbances) and Gaussian noise (each pixel value in the image changes by a small amount). In either case, the noise at different pixels can either be correlated or uncorrelated. In many cases, noise values at different pixels are modeled as being independent and identically distributed and hence uncorrelated. In selecting noise reduction algorithm, one must consider the available computer power and time and whether sacrificing some image detail is acceptable if it allows more noise to be removed and so on.

Deblurring is the process of removing blurring artifacts (such as blur caused by defocus aberration or motion blur) from images. The blur is typically modelled as a convolution point-spread function with a hypothetical sharp input image, where both the sharp input image (which is to be recovered) and the point-spread function are unknown. Deblurring algorithms include methodology to remove the blur from an image. Deblurring is an iterative process

and you might need to repeat the process multiple times until the final image is the best approximation of the original image.

Edge extraction or edge detection is used to separate objects from one another before identifying their contents. It includes a variety of mathematical methods that aim at identifying points in a digital image at which the image brightness changes sharply.

Edge detection approaches can be categorised into search-based approach and zero-crossing-based approach. Search based methods detect edges by first computing a measure of edge strength (usually a first-order derivative function) such as gradient magnitude and then searching for a local directional maxima of the gradient magnitude using a computed estimate of the local orientation of the edge, usually the gradient direction. Zero-crossing methods look for zero-crossings in a second-order derivative function computed from the image to find the edge. First-order edge detectors include Canny edge detector, Prewitt and Sobel operators, and so on.

Other approaches include second-order differential approach of detecting zero-crossings, phase congruency (or phase coherence) methods or phase-stretch transform (PST). Second-order differential approach detects zero-crossings of the second-order directional derivative in the gradient direction. Phase congruency methods attempt to find locations in an image where all sinusoids in the frequency domain are in phase. PST transforms the image by emulating propagation through a diffractive medium with engineered 3D dispersive property (refractive index).

Binarisation refers to reducing a greyscale image to only two levels of grey, i.e., black and white. Thresholding is a popular technique for converting any greyscale image into a binary image.